Novelty Analysis

Analysis of Novelty, Transience, and Resonance

PENPAL Project

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0.1 Summary

- Humans produce higher novelty (surprisal) and higher transience than the model, indicating stronger deviations from earlier story context.
- Both agents display a positive novelty–resonance relationship (innovation bias), but the slope is steeper for humans.
- This indicates that surprising human turns exert greater forward influence on narrative development.
- AI contributions show lower variance and lower informational impact relative to human inputs.
- Humans are the primary drivers of narrative innovation and thematic redirection.

1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

```
# Load Field Data
df_res_field <- read_csv(here("data/TEXT/processed/novelty_scores_40.csv")) %>%
  filter(!is.na(user_surprise) & !is.na(ai_surprise)) %>%
  mutate(
    user_resonance = user_surprise - user_transience,
```

```r
    ai_resonance = ai_surprise - ai_transience,
    datatype = "Field"
  ) %>%
  drop_na(user, ai) %>%
  filter(interaction_count > 1, interaction_count < 10, ai_surprise < 0)
```

1.2 1.2 Reshape Data

```r
# Compute word counts for each agent's turn
df_res_field <- df_res_field %>%
  mutate(
    user_nwords = str_count(user, "\\S+"),
    ai_nwords = str_count(ai, "\\S+")
  )

df_res_long <- df_res_field %>%
  pivot_longer(
    cols = c(user_surprise, user_transience, user_nwords,
             ai_surprise, ai_transience, ai_nwords),
    names_to = c("agent", "metric"),
    names_pattern = "(user|ai)_(surprise|transience|nwords)"
  ) %>%
  pivot_wider(names_from = metric, values_from = value) %>%
  mutate(
    resonance = surprise - transience,
    agent = factor(agent, levels = c("user", "ai"), labels = c("User", "LLM"))
  )
```

1.3 1.3 Descriptive Statistics

```r
desc_stats <- df_res_long %>%
  group_by(agent) %>%
  summarise(
    n = n(),
    mean_surprise = mean(surprise, na.rm = TRUE),
    sd_surprise = sd(surprise, na.rm = TRUE),
    mean_transience = mean(transience, na.rm = TRUE),
    sd_transience = sd(transience, na.rm = TRUE),
    mean_resonance = mean(resonance, na.rm = TRUE),
    sd_resonance = sd(resonance, na.rm = TRUE)
  )

print(desc_stats)
```

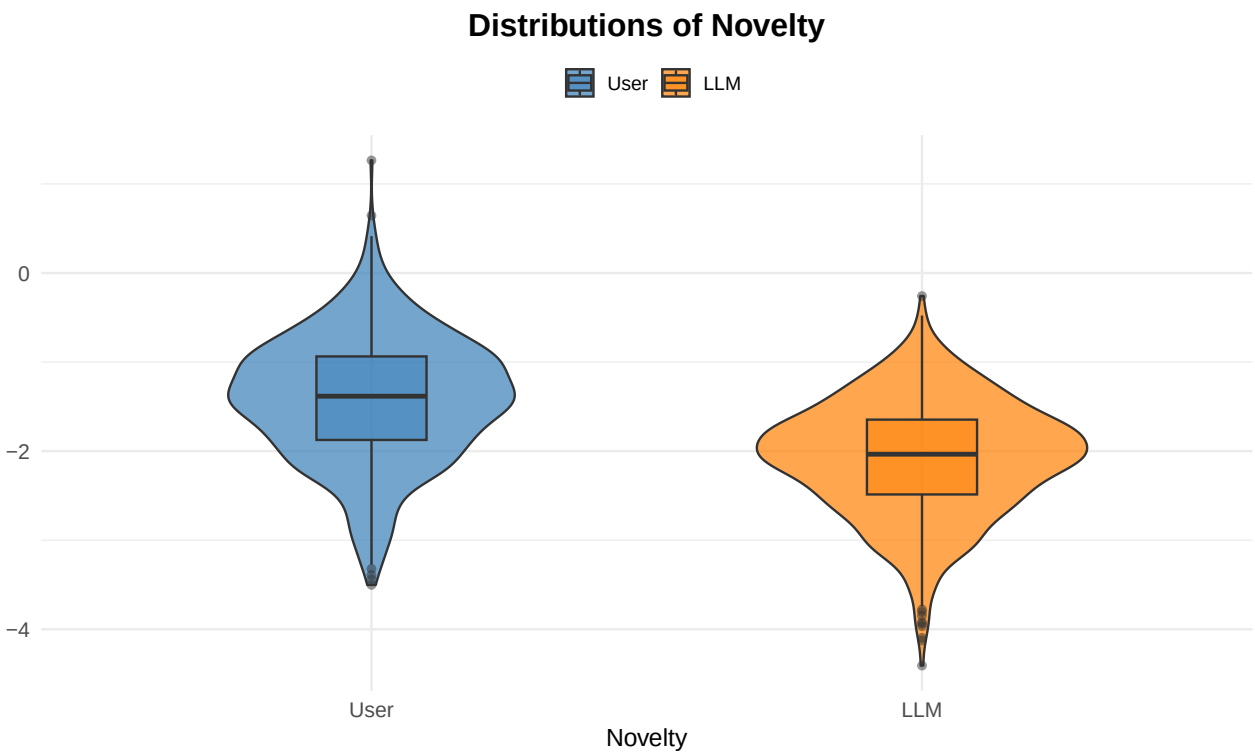
```
## # A tibble: 2 x 8
##   agent     n mean_surprise sd_surprise mean_transience sd_transience
##   <fct> <int>         <dbl>       <dbl>           <dbl>         <dbl>
## 1 User    700         -1.42       0.713          -1.03          0.441
## 2 LLM     700         -2.08       0.647          -0.676         0.454
## # i 2 more variables: mean_resonance <dbl>, sd_resonance <dbl>
```

```r
knitr::kable(desc_stats, caption = "Descriptive Statistics for Novelty Metrics by Agent", digits = 3)
```

Table 1: Descriptive Statistics for Novelty Metrics by Agent

agent	n	mean_surprise	sd_surprise	mean_transience	sd_transience	mean_resonance	sd_resonance
User	700	-1.418	0.713	-1.034	0.441	-0.384	0.774
LLM	700	-2.077	0.647	-0.676	0.454	-1.401	0.684

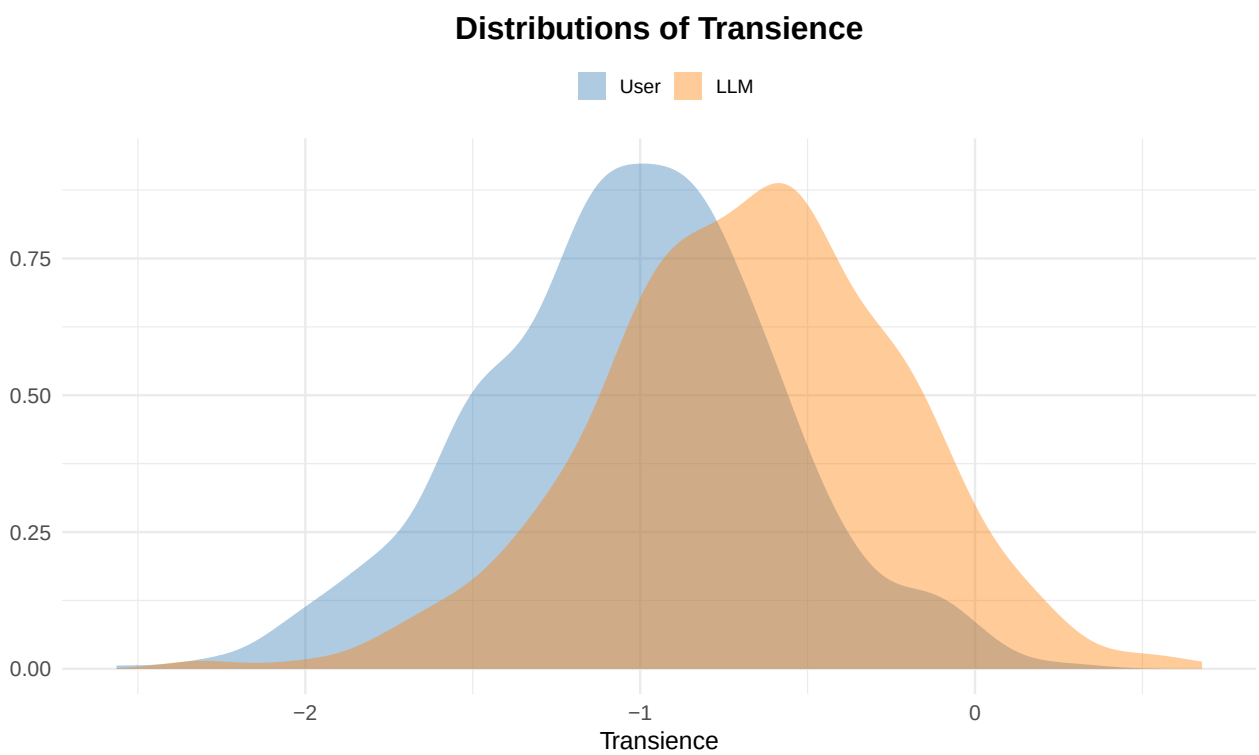
```
p_novelty_agent <- ggplot(df_res_long, aes(x = agent, y = surprise, fill = agent)) +
  geom_violin(width = 0.6, alpha = 0.7) +
  geom_boxplot(width = 0.2, alpha = 0.5) +
  scale_fill_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00"), name = "") +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
  labs(title = "Distributions of Novelty", x = "Novelty", y = NULL) +
  theme(
    legend.position = "top",
    plot.title = element_text(hjust = 0.5, size = 18, face = "bold"),
    axis.text = element_text(size = 12)
  )
print(p_novelty_agent)
```




```
ggsave(
  file.path(GRAPHS_DIR, "Surprise_Distributions_Agent.png"),
  p_novelty_agent,
  width = 10,
  height = 6,
  dpi = 300
)

p_transience_agent <- ggplot(df_res_long, aes(x = transience, fill = agent)) +
  geom_density(alpha = 0.4, color = NA) +
```

```r
  scale_fill_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00"), name = "") +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
  labs(title = "Distributions of Transience", x = "Transience", y = NULL) +
  theme(
    legend.position = "top",
    plot.title = element_text(hjust = 0.5, size = 18, face = "bold"),
    axis.text = element_text(size = 12)
  )
print(p_transience_agent)
```




```r
ggsave(
  file.path(GRAPHS_DIR, "Transience_Distributions_Agent.png"),
  p_transience_agent,
  width = 10,
  height = 6,
  dpi = 300
)
```

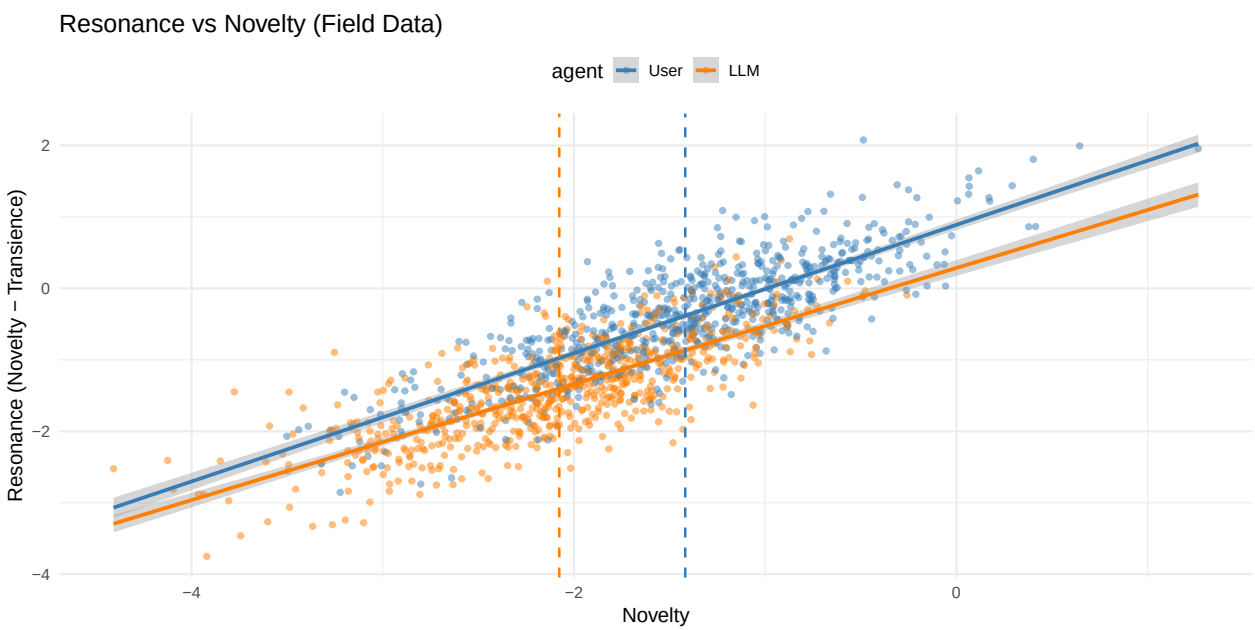
2 2. Novelty and Resonance Analysis

2.1 2.1 Resonance vs Novelty Plot

```r
# Plot Resonance vs Novelty (Field Data)
p_resonance <- ggplot(df_res_long, aes(x = surprise, y = resonance, color = agent)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, fullrange = TRUE, size = 1.2) +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
    geom_vline(data = desc_stats,
               aes(xintercept = mean_surprise, color = agent),
```

```
              linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title = "Resonance vs Novelty (Field Data)",
    x = "Novelty",
    y = "Resonance (Novelty - Transience)"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top")

print(p_resonance)
```

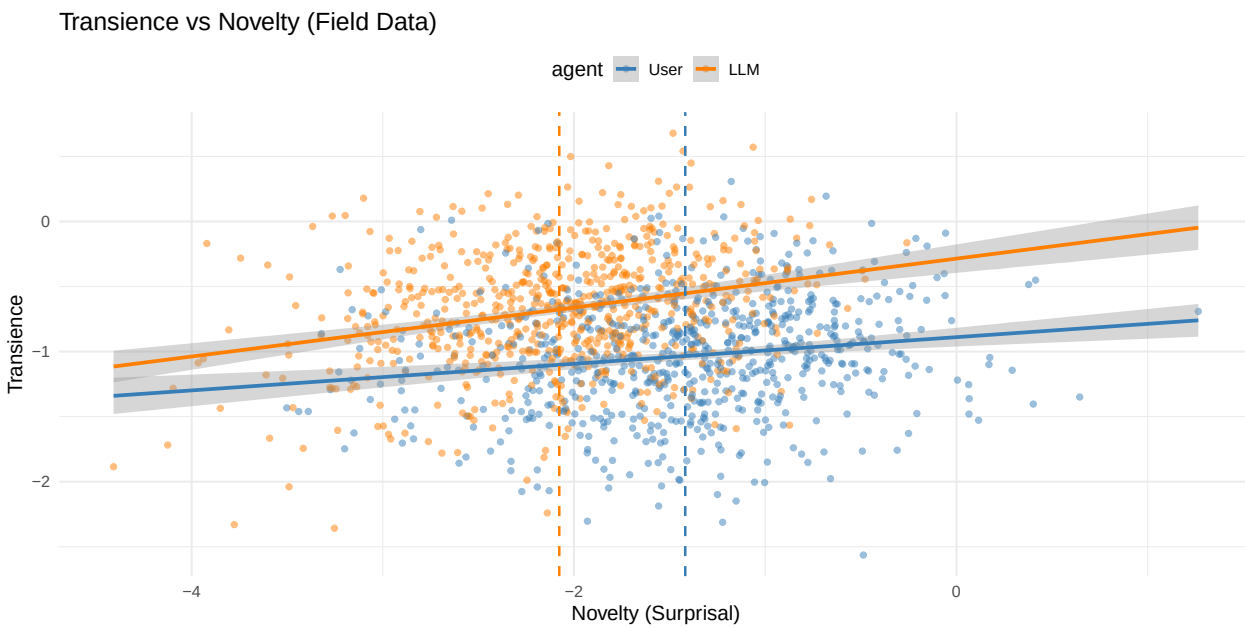



```
ggsave(file.path(GRAPHS_DIR, "novelty_resonance_vs_novelty.png"),
       p_resonance, width = 12, height = 6, dpi = 300)
```

2.2 2.2 Transience vs Novelty Plot

```
p_transience <- ggplot(df_res_long, aes(x = surprise, y = transience, color = agent)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, fullrange = TRUE, size = 1.2) +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
      geom_vline(data = desc_stats,
              aes(xintercept = mean_surprise, color = agent),
              linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title = "Transience vs Novelty (Field Data)",
    x = "Novelty (Surprisal)",
    y = "Transience"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top")

print(p_transience)
```




```
ggsave(file.path(GRAPHS_DIR, "novelty_transience_vs_novelty.png"),
       p_transience, width = 12, height = 6, dpi = 300)
```

3 3. Statistical Models

3.1 3.1 Resonance Model

```
# Model without length control
model_resonance <- lmer(resonance ~ surprise * agent + (1 | conversation_id), data = df_res_long)
cat("=== Resonance Model ===\n")
```

```
## === Resonance Model ===
```

```
print(summary(model_resonance)$coefficients)
```

```
##                     Estimate Std. Error        df   t value      Pr(>|t|)
## (Intercept)        0.9513552 0.03919177  786.1344 24.274361  1.416786e-97
## surprise           0.9414640 0.02273835 1385.4098 41.404236 1.548277e-244
## agentLLM          -0.6145943 0.06511592 1382.0987 -9.438465  1.545908e-20
## surprise:agentLLM -0.1049326 0.03376284 1383.6349 -3.107931  1.922396e-03
```

```
cat("\n=== Slope Comparison (Resonance) ===\n")
```

```
##
## === Slope Comparison (Resonance) ===
```

```
print(emtrends(model_resonance, specs = "agent", var = "surprise"))
```

```
##  agent surprise.trend     SE   df lower.CL upper.CL
##  User           0.941 0.0228 1386    0.897    0.986
##  LLM            0.837 0.0254 1395    0.787    0.886
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```r
# Model WITH length control
model_resonance_len <- lmer(resonance ~ surprise * agent + nwords + (1 | conversation_id), data = df_res
cat("\n=== Resonance Model (controlling for turn length) ===\n")
```

```
##
## === Resonance Model (controlling for turn length) ===
```

```r
print(summary(model_resonance_len)$coefficients)
```

```
##                      Estimate  Std. Error        df    t value      Pr(>|t|)
## (Intercept)        0.74948078 0.070266409  860.0761  10.666274  4.955955e-25
## surprise           0.93477736 0.022727025 1384.4479  41.130653 2.770727e-242
## agentLLM          -0.66658106 0.066625687 1380.6667 -10.004866  8.483003e-23
## nwords             0.00756089 0.002198219  835.0931   3.439552  6.115529e-04
## surprise:agentLLM -0.11480800 0.033768591 1380.9824  -3.399846  6.934649e-04
```

```r
cat("\n=== Slope Comparison (Resonance, length-controlled) ===\n")
```

```
##
## === Slope Comparison (Resonance, length-controlled) ===
```

```r
print(emtrends(model_resonance_len, specs = "agent", var = "surprise"))
```

```
##  agent surprise.trend     SE   df lower.CL upper.CL
##  User           0.935 0.0227 1385    0.890    0.979
##  LLM            0.820 0.0257 1390    0.769    0.870
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

3.2 3.2 Transience Model

```r
# Model without length control
model_transience <- lmer(transience ~ surprise * agent + (1 | conversation_id), data = df_res_long)

cat("=== Transience Model ===\n")
```

```
## === Transience Model ===
```

```r
print(summary(model_transience)$coefficients)
```

```
##                      Estimate Std. Error        df    t value     Pr(>|t|)
## (Intercept)       -0.95135519 0.03919177  786.1344 -24.274361 1.416785e-97
## surprise           0.05853604 0.02273835 1385.4098   2.574331 1.014649e-02
## agentLLM           0.61459428 0.06511592 1382.0987   9.438465 1.545908e-20
## surprise:agentLLM  0.10493258 0.03376284 1383.6349   3.107931 1.922396e-03
```

```r
cat("\n=== Slope Comparison (Transience) ===\n")
```

```
##
## === Slope Comparison (Transience) ===
```

```r
print(emtrends(model_transience, specs = "agent", var = "surprise"))
```

```
##  agent surprise.trend     SE   df lower.CL upper.CL
##  User          0.0585 0.0228 1386   0.0139    0.103
##  LLM           0.1635 0.0254 1395   0.1136    0.213
##
```

```
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```r
# Model WITH length control
model_transience_len <- lmer(transience ~ surprise * agent + nwords + (1 | conversation_id), data = df_

cat("\n=== Transience Model (controlling for turn length) ===\n")
```

```
##
## === Transience Model (controlling for turn length) ===
```

```r
print(summary(model_transience_len)$coefficients)
```

```
##                      Estimate  Std. Error        df    t value     Pr(>|t|)
## (Intercept)       -0.74948078 0.070266409  860.0761 -10.666274 4.955956e-25
## surprise           0.06522264 0.022727025 1384.4479   2.869827 4.169357e-03
## agentLLM           0.66658106 0.066625687 1380.6667  10.004866 8.483002e-23
## nwords            -0.00756089 0.002198219  835.0931  -3.439552 6.115529e-04
## surprise:agentLLM  0.11480800 0.033768591 1380.9824   3.399846 6.934649e-04
```

```r
cat("\n=== Slope Comparison (Transience, length-controlled) ===\n")
```

```
##
## === Slope Comparison (Transience, length-controlled) ===
```

```r
print(emtrends(model_transience_len, specs = "agent", var = "surprise"))
```

```
##  agent surprise.trend     SE   df lower.CL upper.CL
##  User          0.0652 0.0227 1385   0.0206    0.110
##  LLM           0.1800 0.0257 1390   0.1295    0.231
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

4 4. Supplementary: Resonance vs Novelty by LLM Type

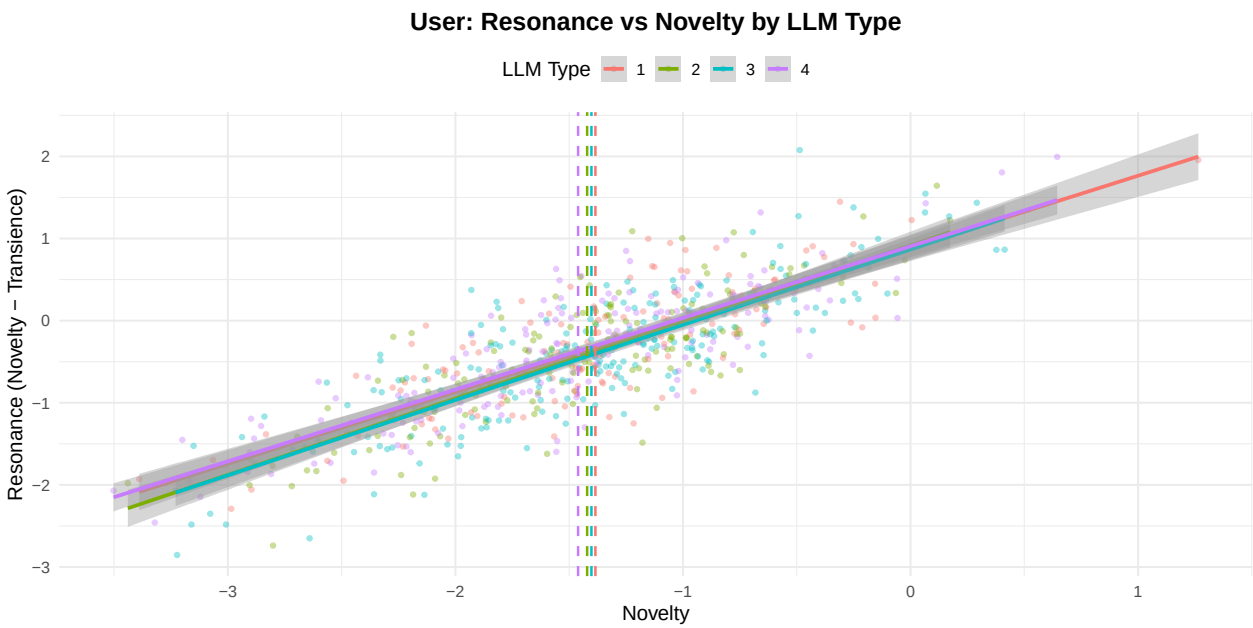
```r
desc_stats_llm <- df_res_long %>%
  filter(!is.na(llm_type)) %>%
  group_by(agent, llm_type) %>%
  summarise(mean_surprise = mean(surprise, na.rm = TRUE), .groups = "drop")

p_llm_user <- df_res_long %>%
  filter(!is.na(llm_type), agent == "User") %>%
  ggplot(aes(x = surprise, y = resonance, color = factor(llm_type))) +
  geom_point(alpha = 0.4, size = 1.2) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.2) +
  geom_vline(data = desc_stats_llm %>% filter(agent == "User"),
             aes(xintercept = mean_surprise, color = factor(llm_type)),
             linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title  = "User: Resonance vs Novelty by LLM Type",
    x      = "Novelty",
    y      = "Resonance (Novelty - Transience)",
    color  = "LLM Type"
  ) +
  theme_minimal(base_size = 14) +
```

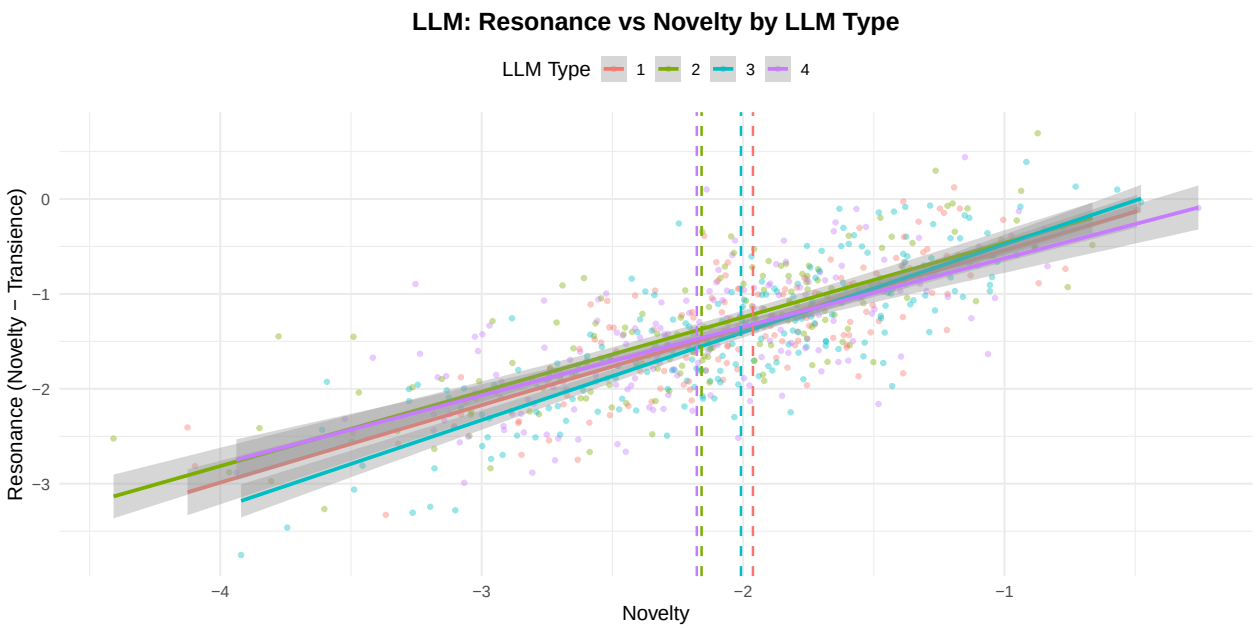
```r
  theme(legend.position = "top",
        plot.title = element_text(hjust = 0.5, face = "bold"))

p_llm_ai <- df_res_long %>%
  filter(!is.na(llm_type), agent == "LLM") %>%
  ggplot(aes(x = surprise, y = resonance, color = factor(llm_type))) +
  geom_point(alpha = 0.4, size = 1.2) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.2) +
  geom_vline(data = desc_stats_llm %>% filter(agent == "LLM"),
             aes(xintercept = mean_surprise, color = factor(llm_type)),
             linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title  = "LLM: Resonance vs Novelty by LLM Type",
    x      = "Novelty",
    y      = "Resonance (Novelty - Transience)",
    color  = "LLM Type"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top",
        plot.title = element_text(hjust = 0.5, face = "bold"))

print(p_llm_user)
```




```r
print(p_llm_ai)
```




```r
ggsave(file.path(GRAPHS_DIR, "novelty_resonance_llm_type_user.png"),
       p_llm_user, width = 12, height = 6, dpi = 300)
ggsave(file.path(GRAPHS_DIR, "novelty_resonance_llm_type_ai.png"),
       p_llm_ai, width = 12, height = 6, dpi = 300)
```

---

5 5. Session Information

```r
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.3
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK ve
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Australia/Melbourne
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_2.0.1   here_1.0.2      afex_1.4-1      cowplot_1.1.3
##  [5] patchwork_1.3.0 zoo_1.8-12      stringi_1.8.4   arrow_19.0.1.1
##  [9] lmerTest_3.1-3  lme4_1.1-35.5   Matrix_1.7-1    lubridate_1.9.3
## [13] forcats_1.0.0   stringr_1.5.1   dplyr_1.1.4     purrr_1.0.2
```

```
## [17] readr_2.1.5     tidyr_1.3.1     tibble_3.2.1    ggplot2_4.0.2
## [21] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    farver_2.1.2        S7_0.2.1
##  [4] fastmap_1.2.0       TH.data_1.1-2       pacman_0.5.1
##  [7] digest_0.6.37       timechange_0.3.0    estimability_1.5.1
## [10] lifecycle_1.0.5     survival_3.7-0      magrittr_2.0.3
## [13] compiler_4.4.2      rlang_1.1.7         tools_4.4.2
## [16] utf8_1.2.4          yaml_2.3.10         knitr_1.50
## [19] labeling_0.4.3      bit_4.0.5           plyr_1.8.9
## [22] RColorBrewer_1.1-3  abind_1.4-8         multcomp_1.4-26
## [25] withr_3.0.2         numDeriv_2016.8-1.1 grid_4.4.2
## [28] fansi_1.0.6         xtable_1.8-4        scales_1.4.0
## [31] MASS_7.3-61         tinytex_0.52        cli_3.6.5
## [34] mvtnorm_1.3-3       rmarkdown_2.28      crayon_1.5.3
## [37] ragg_1.3.2          generics_0.1.3      rstudioapi_0.16.0
## [40] reshape2_1.4.4      tzdb_0.4.0          minqa_1.2.8
## [43] splines_4.4.2       assertthat_0.2.1    parallel_4.4.2
## [46] vctrs_0.7.1         boot_1.3-31         sandwich_3.1-1
## [49] carData_3.0-5       car_3.1-3           hms_1.1.3
## [52] pbkrtest_0.5.3      bit64_4.0.5         Formula_1.2-5
## [55] systemfonts_1.1.0   glue_1.8.0          nloptr_2.1.1
## [58] codetools_0.2-20    gtable_0.3.6        pillar_1.9.0
## [61] htmltools_0.5.8.1   R6_2.6.1            textshaping_0.4.0
## [64] rprojroot_2.1.1     vroom_1.6.5         evaluate_1.0.3
## [67] lattice_0.22-6      backports_1.5.0     broom_1.0.6
## [70] Rcpp_1.0.13         coda_0.19-4.1       nlme_3.1-166
## [73] mgcv_1.9-1          xfun_0.54           pkgconfig_2.0.3
```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.